

ESG Token Automation Tool

Automated E-Token generation for EastStagingGrounds (eaststaginggrounds.com.sg)

Installation (End Users)

No installation required. Simply:

1. Unzip the `dist` folder you received
2. Double-click `ESG_Automation.exe`
3. Your browser will open automatically at `http://127.0.0.1:5000`

Requirements: Windows 10 or later (64-bit). Nothing else needed — Python, Chrome, and all dependencies are bundled inside.

Quick Start

1. Launch `ESG_Automation.exe`
2. Open your browser and go to `http://127.0.0.1:5000`
3. Add your account(s) in the **Accounts** section
4. Upload an Excel file or enter truck data manually
5. (Optional) Set a **Scheduled Start Time**
6. Click **Start Automation**

Features

1. Multi-Account Support

- Add as many accounts as you need in the Accounts panel
- Each account runs in its own browser session **simultaneously**
- All sessions log in and navigate to the token generation page in parallel

2. Automatic Load Balancing

- You only need **one master list** of trucks
- The system automatically splits the list evenly across all your accounts
- **Example:** 30 trucks + 3 accounts → Account A gets 1–10, Account B gets 11–20, Account C gets 21–30
- No more managing multiple Excel files

3. Excel Upload

- Upload a `.xlsx` file with your truck data
- Required columns: **Truck No**, **Material Type**, **Source Site**, **Staging Ground**
- The file is parsed instantly and displayed in the preview table

4. Manual Truck Entry

- Toggle to **Manual Entry** mode if you don't have an Excel file
- Add and remove rows directly in the browser UI
- Fill in Truck No, Material, Source Site, and Staging Ground for each row

5. Scheduled Start Timer (Synchronized)

- Enter a target time in `HH:MM:SS` format (e.g., `08:00:00`)
- The bot will:
- Immediately launch all browsers and log in to all accounts
- Navigate every account to the token generation page
- **Wait** on that page until the exact scheduled time
- **Blast** all token generation requests simultaneously the moment the time hits
- **External Time Sync:** The system fetches the true global time from Google's servers at startup to correct for any local clock drift, ensuring tokens are generated at the exact right millisecond

6. Configurable Wait Time

- Set the **Wait Before Generation (ms)** to control the delay between clicking "Go" and clicking "Generate"
- Default: **50ms** (very fast)
- Increase to **300–500ms** if you see `"Please click Go before generate E-token"` errors, which means the server needs more time to validate the truck before the token can be generated

7. Live Log Stream

- All activity from all accounts is streamed live to the dashboard

- Each log line is prefixed with the account username so you can track each session independently

8. Live Results Table

- The results table updates in real-time as each truck is processed
- Status column shows  **Success** or  **Error** with the exact message from the website

9. Download Report (CSV)

- After automation finishes, a **Download Report (CSV)** button appears
- Click it to instantly download all results as a `.csv` file
- The report includes: Index, Account, Truck No, Material, Status, and Error Message

10. Stop Automation

- Click the **Stop** button at any time to gracefully halt all running sessions
- The browser will close cleanly after the current row finishes

Excel File Format

Your Excel file must have these column headers (exact names):

Column	Description	Example
Truck No	Vehicle plate number	XD9754T
Material Type	Type of material	GOOD EARTH
Source Site	Source site name	TMSGHPCE027
Staging Ground	Target staging ground	TMSG

Adding Accounts

1. In the **Accounts** section on the dashboard, click **+ Add Account**
2. Enter the **Username** and **Password** for each account
3. Add as many accounts as needed — there is no limit
4. The load balancer will distribute trucks evenly among all configured accounts

Scheduled Timer Guide

The scheduled timer is designed for high-volume token generation windows that open at a specific time (e.g., 8:00 AM).

Recommended workflow: 1. Prepare your truck list the night before 2. Launch the EXE and configure everything 3. Set the scheduled time to e.g. `08:00:00` 4. Click **Start** — the bots log in immediately and sit ready on the token generation page 5. At exactly 08:00:00 (true global time), all accounts fire simultaneously

Why external time sync? Your computer's clock may run slightly slow. Even a 1–2 second delay means fewer tokens generated before the window closes. The system queries Google's servers the moment you click Start and automatically compensates for clock drift.

For Developers

Running from Source

```
# Install dependencies
pip install flask playwright pandas requests

# Install browser
playwright install chromium

# Run
python app.py
```

Then open `http://127.0.0.1:5000` in your browser.

Building the EXE

```
python build_exe.py
```

This will: 1. Install all required packages 2. Install the Chromium browser 3. Compile `app.py` and all dependencies into `dist/ESG_Automation.exe` 4. Copy the Chromium browser into `dist/ms-playwright`

The final `dist/` folder is fully portable — zip it and distribute it.

Project Structure

```
eaststaginggrounds.com.sg/
├─ app.py                # Flask backend, API endpoints, thread pool
```

```

├── automation.py          # Playwright browser automation logic
├── build_exe.py           # PyInstaller build script
├── hook_fix_pkgresources.py # PyInstaller runtime patch
├── app_icon.ico           # Application icon
├── templates/
│   └── dashboard.html     # Frontend UI (single page)
├── uploads/               # Temporary Excel upload storage
└── dist/                  # Built executable output
    ├── ESG_Automation.exe
    └── ms-playwright/     # Bundled Chromium browser

```

? Common Errors

Error Message	Cause	Fix
Please click Go before generate E-token	Wait time too short — server hasn't validated the truck yet	Increase Wait Before Generation to 300–500ms
VTs Truck data not found for source site	The truck plate number is not registered in the system	Verify the truck number and source site in the website
Timeout exceeded	Page took too long to load	Check your internet connection; the site may be slow
Browser window not opening	Antivirus blocking the bundled Chromium	Add the dist folder to your antivirus exclusion list

📞 Support

For issues or feature requests, contact your system administrator.